Agrupando com GROUP BY

```
SELECT gp.ref_year,
       AVG(gp.gdp_pc) AS avg_gdp_pc
FROM gdp_pc gp
JOIN country c ON gp.country = c.country
WHERE gp.ref_year BETWEEN 2018 AND 2022
  AND c.four_regions = 'americas'
GROUP BY gp.ref_year;
```

	123 ref_year	123 avg_gdp_pc
1	2,018	18,167.4285714286
2	2,019	18,220.2857142857
3	2,020	16,484.8571428571
4	2,021	17,536.2857142857
5	2,022	18,682

Agrupando por mais de uma coluna

```
SELECT c.four_regions,
       gp.ref_year,
       ROUND(AVG(gp.gdp_pc), 2) AS avg_gdp_pc
FROM gdp_pc gp
JOIN country c ON gp.country = c.country
WHERE gp.ref_year BETWEEN 2018 AND 2022
GROUP BY c.four_regions,
         gp.ref_year
ORDER BY c.four_regions,
         gp.ref_year;
```

Agrupando por mais de uma coluna – resultado

	four_regions	ref_year	avg_gdp_pc
1	africa	2,018	5,765.09
2	africa	2,019	5,746.19
3	africa	2,020	5,259.36
4	africa	2,021	5,473.36
5	africa	2,022	5,632.51
6	americas	2,018	18,167.43
7	americas	2,019	18,220.29
8	americas	2,020	16,484.86
9	americas	2,021	17,536.29
10	americas	2,022	18,682
11	asia	2,018	20,645.76
12	asia	2,019	20,860.17
13	asia	2,020	19,924.92
14	asia	2,021	20,745.25
15	asia	2,022	21,306.44
16	europe	2,018	42,595.65
17	europe	2,019	43,450
18	europe	2,020	41,680.43
19	europe	2,021	44,113.04
20	europe	2,022	45,610.87

Selecionando por valor agrupado com HAVING

```
SELECT c.four_regions,
       gp.ref_year,
       ROUND(AVG(gp.gdp_pc), 2) AS avg_gdp_pc
FROM gdp_pc gp
JOIN country c ON gp.country = c.country
WHERE gp.ref_year BETWEEN 2018 AND 2022
GROUP BY c.four_regions,
         gp.ref_year
HAVING avg_gdp_pc > 17000
ORDER BY c.four_regions,
         gp.ref_year;
```

Selecionando por valor agrupado com HAVING – resultado

	four_regions	ref_year	avg_gdp_pc
1	americas	2,018	18,167.43
2	americas	2,019	18,220.29
3	americas	2,021	17,536.29
4	americas	2,022	18,682
5	asia	2,018	20,645.76
6	asia	2,019	20,860.17
7	asia	2,020	19,924.92
8	asia	2,021	20,745.25
9	asia	2,022	21,306.44
10	europe	2,018	42,595.65
11	europe	2,019	43,450
12	europe	2,020	41,680.43
13	europe	2,021	44,113.04
14	europe	2,022	45,610.87

HAVING por faixa de valores

```
SELECT c.four_regions,
       gp.ref_year,
       ROUND(AVG(gp.gdp_pc), 2) AS avg_gdp_pc
FROM gdp_pc gp
JOIN country c ON gp.country = c.country
WHERE gp.ref_year BETWEEN 2018 AND 2022
GROUP BY c.four_regions,
         gp.ref_year
HAVING avg_gdp_pc BETWEEN 17000 AND 20000
ORDER BY c.four_regions,
         gp.ref_year;
```

	four_regions	ref_year	avg_gdp_pc
1	americas	2,018	18,167.43
2	americas	2,019	18,220.29
3	americas	2,021	17,536.29
4	americas	2,022	18,682
5	asia	2,020	19,924.92

Negando o HAVING

```
SELECT c.four_regions,
       gp.ref_year,
       ROUND(AVG(gp.gdp_pc), 2) AS avg_gdp_pc
FROM gdp_pc gp
JOIN country c ON gp.country = c.country
WHERE gp.ref_year BETWEEN 2018 AND 2022
GROUP BY c.four_regions,
         gp.ref_year
HAVING NOT avg_gdp_pc > 17000
ORDER BY c.four_regions,
         gp.ref_year;
```

	four_regions	ref_year	avg_gdp_pc
1	africa	2,018	5,765.09
2	africa	2,019	5,746.19
3	africa	2,020	5,259.36
4	africa	2,021	5,473.36
5	africa	2,022	5,632.51
6	americas	2,020	16,484.86

Contando as linhas de uma tabela com COUNT()

```
SELECT count(*)
FROM women_years_at_school;
```

	123 count(*)
1	7,000

Contando valores distintos com COUNT(DISTINCT)

```
SELECT count(DISTINCT country)
FROM women_years_at_school;
```

	123 count(DISTINCT country)
1	175

```
SELECT count(DISTINCT ref_year)
FROM women_years_at_school;
```

	123 count(DISTINCT ref_year)
1	40

Usando outras funções com o COUNT(DISTINCT)

```
SELECT count(DISTINCT ref_year),
    min(ref_year),
    max(ref_year)
FROM women_years_at_school;
```

	123 count(DISTINCT ref_year)	123 min(ref_year)	123 max(ref_year)
1	40	1,970	2,009

Contando com GROUP BY

```
SELECT country,
    count(*)
FROM women_years_at_school
GROUP BY country;
```

	country	count(*)
1	Afghanistan	40
2	Albania	40
3	Algeria	40
4	Angola	40
5	Antigua and Barbuda	40
6	Argentina	40
7	Armenia	40
8	Australia	40
9	Austria	40
10	Azerbaijan	40

Contando com GROUP BY – outro enfoque

```
SELECT ref_year,
    count(*)
FROM women_years_at_school
GROUP BY ref_year;
```

	123 ref_year	123 count(*)
1	1,970	175
2	1,971	175
3	1,972	175
4	1,973	175
5	1,974	175
6	1,975	175
7	1,976	175
8	1,977	175
9	1,978	175
10	1,979	175

Contando valores nulos

```
SELECT count(*)
FROM women_years_at_school
WHERE mean_years IS NULL;
```

	123 count(*)
1	0

FGV EESP

Todos os direitos reservados. Textos, vídeos, sons, imagens, gráficos e demais componentes deste material são protegidos por direitos autorais e outros direitos de propriedade intelectual, de forma que é proibida a reprodução no todo ou em parte, sem a devida autorização.

Como citar este material:
BONATO, Antonio Sergio Ferreira. **Agrupando dados**. Rio de Janeiro: FGV, 2024.